INCH-POUND
A-A-59569
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

# BRAID, WIRE (COPPER, TIN-COATED OR SILVER-COATED, TUBULAR OR FLAT)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This CID covers tin- or silver-coated copper wires braided in tubular or flat form intended for use as shielding over electrical conductors or connections to motor brushes, controller contacts, and grounding bonds.

2. SALIENT CHARACTERISTICS.

2.1 Materials. The wire used in the construction of braids shall conform to ASTM B33 for tinned copper and to ASTM B298 for silver-coated copper, except that the silver coating shall be a minimum of 40 micro-inches thick.

2.2 Design and construction. The wire braid shall have the numbers of carriers and ends shown in table I.

2.2.1 Wire lengths. Individual wires shall be to sizes shown in table I and shall be uniform in cross-section. Each wire shall be one continuous length, free from splices except as specified herein.

2.2.2 Splices.

2.2.2.1 Carrier splices. There shall be no more than one splice or break in any carrier in each 25-foot length of the braid.

2.2.2.2 Wire splices. Excluding the carrier splice, there shall be no more than one broken or spliced end of wire in each 25-foot length of the braid.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data that may improve this document should be sent to: Defense Logistics Agency, Defense Supply Center, Columbus (DSCC-VAI), P.O. Box 3990, Columbus, OH 43216-5000.

FSC 6145

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TABLE I. Braid dimensions and data.

| PIN 1/ AA59569 | Strand wire size (AWG) | Tubular inside diameter (inches) 2/ | Number of carriers | Number of ends | Current rating (amps) 3/ | Approx. AWG equiv. 4/ | Flat form width x thickness (inches) 5/ |
|---|---|---|---|---|---|---|---|
| *36*0031 | 36 | .031 | 24 | 24 | 7.0 | 22 | .046 x .020 |
| *36*0062 | 36 | .062 | 24 | 48 | 11.0 | 19 | .093 x .031 |
| *34*0062 | 34 | .062 | 16 | 32 | 11.0 | 19 | - |
| *32*0062 | 32 | .062 | 16 | 16 | 9.0 | 20 | - |
| *36*0078 | 36 | .078 | 24 | 72 | 16.0 | 18 | .125 x .020 |
| *36*0109 | 36 | .109 | 24 | 96 | 19.0 | 16 | .156 x .031 |
| *34*0109 | 34 | .109 | 16 | 64 | 19.0 | 16 | - |
| *32*0109 | 32 | .109 | 16 | 32 | 18.0 | 17 | - |
| *36*0125 | 36 | .125 | 24 | 120 | 25.0 | 15 | .187 x .020 |
| *34*0125 | 34 | .125 | 24 | 72 | 19.0 | 16 | - |
| *32*0125 | 32 | .125 | 24 | 48 | 25.0 | 15 | - |
| *36*0156 | 36 | .156 | 24 | 240 | 40.0 | 12 | .250 x .046 |
| *36*0171 | 36 | .171 | 24 | 168 | 32.0 | 14 | .250 x .030 |
| *34*0171 | 34 | .171 | 24 | 120 | 36.0 | 13 | - |
| *32*0171 | 32 | .171 | 24 | 72 | 32.0 | 14 | - |
| *36*0203 | 36 | .203 | 24 | 312 | 46.0 | 11 | .281 x .046 |
| *34*0203 | 34 | .203 | 24 | 192 | 46.0 | 11 | - |
| *32*0203 | 32 | .203 | 24 | 120 | 46.0 | 11 | - |
| *36*0250 | 36 | .250 | 24 | 384 | 53.0 | 10 | - |
| *30*0281 | 30 | .281 | 24 | 120 | 60.0 | 9 | - |
| *36*0375 | 36 | .375 | 48 | 384 | 53.0 | 10 | .625 x .030 |
| *34*0375 | 34 | .375 | 48 | 240 | 53.0 | 10 | - |
| *32*0375 | 32 | .375 | 48 | 144 | 46.0 | 11 | - |
| *30*0375 | 30 | .375 | 24 | 168 6/ | 75.0 | 8 | - |
| *30*0437 | 30 | .437 | 24 | 240 | 90.0 | 6 | .500 x .093 |
| *36*0500 | 36 | .500 | 48 | 528 | 62.0 | 9 | .625 x .046 |
| *34*0500 | 34 | .500 | 48 | 336 | 62.0 | 9 | - |
| *32*0500 | 32 | .500 | 48 | 192 | 62.0 | 9 | - |
| *30*0500 | 30 | .500 | 24 | 360 | 120.0 | 6 | .625 x .093 |
| *30*0562 | 30 | .562 | 48 | 480 | 145.0 | 3 | - |
| *30*0656 | 30 | .656 | 48 | 768 | 190.0 | 1 | - |
| *36*0781 | 36 | .781 | 48 | 864 | 88.0 | 7 | .750 x .040 |
| *34*0781 | 34 | .781 | 48 | 528 | 88.0 | 7 | - |
| *32*0781 | 32 | .781 | 48 | 336 | 88.0 | 7 | - |
| *30*0875 | 30 | .875 | 48 | 336 | 100.0 | 5 | 1.375 x .050 |
| *30*1000 | 30 | 1.000 | 48 | 384 | 120.0 | 4 | - |
| *30*1125 | 30 | 1.125 | 48 | 432 | 130.0 | 4 | - |
| *30*1375 | 30 | 1.375 | 48 | 528 | 150.0 | 3 | 1.500 x .060 |

NOTES:

1/ The complete part or identifying number (PIN) shall include additional information to indicate the form (first asterisk) and strand coating (second asterisk) (see 6.1).

2/ Dimensional tolerances shall be as shown in table II.

3/ Direct current ratings are given for information only and are not requirements. Values shown are for uninsulated braid in free air at 30 °C. Values should be derated if the braid is insulated or in close contact with other components.

4/ Approximate AWG equivalents are given for information only and are not requirements.

5/ Flat form width and thickness are given for information only and are not requirements. Tolerances shall be as shown in table II.

6/ This PIN supersedes the similar construction using 96 ends (NSN 6145-00-191-8402).



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TABLE II. Dimensional tolerances (inches).

| Dimensions | Tolerance |
|---|---|
| .000 - .099 | .010 |
| .100 - .249 | .016 |
| .250 - .499 | .031 |
| .500 - .999 | .063 |
| Over .999 | .094 |

2.2.3 Coverage (tubular braid only). Tubular braids shall have a braid angle or number of picks per inch that produces a minimum of 90 percent coverage, except for .078-inch and smaller diameter braids, which shall produce a minimum of 70 percent coverage. The percent of coverage shall be determined by using the following formula.

$$K = 100\left( 2F - F^{2} \right)$$

$$F = \frac{NPW}{CSinA}$$

$$\tan A = \frac{2\ (D + 2W)P}{C}$$

Where:

K = percent coverage of braided shield
A = braid angle
C = number of carriers (table I)
D = inside diameter in inches (table I)
N = total number of ends (table I)
P = picks per inch (the number of times the carriers in a braid cross over each other in the same direction along the longitudinal axis for each inch of length)
W= diameter of individual braid wire in inches

2.3 Flattening of tubular braid. Unless otherwise specified in the contract or order (see 6.2), tubular shielding braid shall not be flattened beyond the point that would occur by its own weight only, when wound on spools for shipping.

2.4 Workmanship. Braids shall be free from lumps, kinks, splits, abrasions, scraped or corroded surfaces, and skin impurities. Braids shall possess good electrical and mechanical solder joints when tested in accordance with MIL-STD-202, method 208.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

4.2 Market acceptability. The item offered must have been sold to the Government or commercial market.



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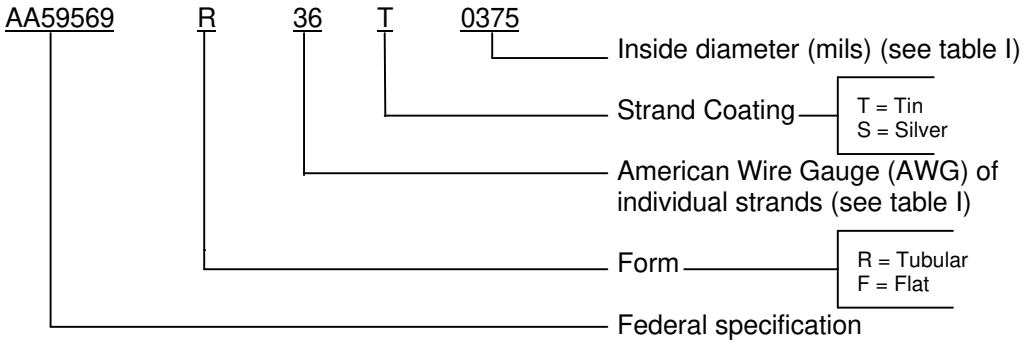
5. PACKAGING. Packing, packaging, and marking shall be as specified in the contract or order (see 6.2 and 6.4).

6. NOTES.

6.1 Part or Identification Number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

The PIN is made up of the basic CID number and a four-element alphanumeric number (see table I):

Example of PIN: AA59569R36T0375




6.2 Ordering data. The contract or order should specify the following:

a. Number, title, and date of this CID
b. PIN (see table I and 6.1)
c. Resistance to flattening, if other than specified (see 2.3)
d. Packaging requirements (see 5 and 6.4)
e. Unspliced braid length requirements and proportions (see 6.3)

6.3 Braid Lengths. Unless otherwise specified in the contract or order (see 6.2), 90 percent of the total order for braid shall be supplied in continuous, unspliced lengths as shown in table III. The remaining 10 percent of the total order for braid shall be acceptable in random unspliced lengths as shown in table III.

TABLE III. Unspliced braid length.

| Braid Diameter (inches) | Unspliced lengths | |
|---|---|---|
| | 90% (nominal lengths) | 10% (minimum lengths) |
| ≤ .171 | 250 feet | 50 feet |
| ≥ .203 | 100 feet | 25 feet |

6.4 Packaging requirements. The packaging and marking requirements for wire braid should be in accordance with MIL-C-12000, except for the marking of shipments to civil agencies and the additional marking of spools and reels, both listed below.

6.4.1 Marking for civil agency procurement. The container marking of shipments to civil agencies should be in accordance with FED-STD-123.



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6.4.2 Marking of spools and reels. In addition to the marking requirements of MIL-C-12000, each spool and reel should be marked with the following information:

a. PIN
b. Net weight (in pounds)
c. Date (month, day, and year) of manufacture
d. Manufacturer's CAGE Code or trademark

6.5 Source of documents.

6.5.1 ASTM standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

| | | |
|---|---|---|
| ASTM B33 | - | Tinned Soft or Annealed Copper Wire for Electrical Purposes (DoD adopted) |
| ASTM B298 | - | Silver-Coated Soft or Annealed Copper Wire (DoD adopted) |

6.5.2 Copies of federal and military specifications and standards required by contractors in connection with specific procurement functions are obtained from the Document Automation and Production Service, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

| | | |
|---|---|---|
| FED-STD-123 | - | Marking for Shipment (Civil Agencies) |
| MIL-STD-202 | - | Test Methods for Electronic and Electrical Component Parts |
| MIL-C-12000 | - | Cable, Cord, and Wire, Electric, Packaging of |

6.5.3 The Federal Acquisition Regulation (FAR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

6.6 Key words.

Wire braid
Shield, braid

MILITARY INTERESTS:

Custodians:
Army - CR
Navy - AS
Air Force - 11
DLA - CC

Review Activities:
Army - AR, AT, AV, MI
Navy - MC, OS, SH
DTRA - DS
NSA - NS

CIVIL AGENCY
COORDINATING ACTIVITIES:

GSA - FSS

Preparing Activity:
DLA - CC

(Project 6145-2286)



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